

**CADANGAN
JAWAPAN
PSPM SEMESTER 2
SK025
SESI 2022/2023**

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Answer **all** questions.

1. (a) An aqueous solution of chlorine dioxide undergoes the following reaction in an alkaline solution.



TABLE 1 shows the results of the initial rate for the reaction at different initial concentrations of ClO_2 and OH^- .

TABLE 1

Experiment	$[\text{ClO}_2] / \text{mol L}^{-1}$	$[\text{OH}^-] / \text{mol L}^{-1}$	Initial rate / $\text{mol L}^{-1} \text{s}^{-1}$
1	1.25×10^{-2}	1.25×10^{-2}	2.33×10^{-4}
2	2.50×10^{-2}	1.25×10^{-2}	9.32×10^{-4}
3	5.00×10^{-2}	2.50×10^{-2}	x

- (i) If the rate constant, k , is $119.3 \text{ L}^2 \text{mol}^{-2} \text{s}^{-1}$, determine the rate law of the reaction.
- (ii) Calculate the initial rate, x in the experiment above.

[6 marks]

- (b) Thermal decomposition of phosphorus pentachloride, PCl_5 at a certain temperature is shown below:



The time taken for 1.00 M of PCl_5 to decrease by 50% is 80 minutes and it takes twice as much time to decrease by another 50%. Deduce the rate constant, k .

[4 marks]

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[illegible]

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2. (a) Hydrazine, N_2H_4 is a colourless flammable liquid and with ammonia-like odour. The combustion of liquid N_2H_4 produces water and nitrogen gas. By using the data given in **TABLE 2.1**, calculate the standard enthalpy of formation of liquid N_2H_4 .

TABLE 2.1

Enthalpy changes	$\Delta H^\circ / \text{kJ mol}^{-1}$
Standard enthalpy of combustion of $\text{N}_2\text{H}_4 (l)$	-623.0
Standard enthalpy of formation of $\text{H}_2\text{O} (l)$	-285.5

[3 marks]

- (b) **TABLE 2.2** shows several values of standard enthalpy changes.

TABLE 2.2

Enthalpy changes	$\Delta H^\circ / \text{kJ mol}^{-1}$
Enthalpy of atomisation of chlorine	+121.0
Electron affinity of chlorine	-364.0
Enthalpy of sublimation of zinc	+126.0
First ionisation energy of zinc	+906.0
Second ionisation energy of zinc	+1733.0
Enthalpy of formation of zinc chloride	-416.0

- (i) Construct the Born-Haber cycle.
(ii) Determine the lattice energy of zinc chloride.
(iii) Compare the lattice energy of zinc chloride with potassium chloride. Explain your answer.

[7 marks]

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NO	PART	SCHEME	MARKS	
2	(a)	$2\text{H}_2\text{O}(l) + \text{N}_2(g) \rightarrow \text{N}_2\text{H}_4(l) + \text{O}_2(g) \quad \Delta H = + 623 \text{ kJ mol}^{-1}$	1	M1
		$2\text{H}_2(g) + \text{O}_2(g) \rightarrow 2\text{H}_2\text{O}(l) \quad \Delta H = (-285.5 \times 2) \text{ kJ mol}^{-1}$	1	M2
		$2\text{H}_2(g) + \text{N}_2(g) \rightarrow \text{N}_2\text{H}_4(l) \quad \Delta H = 623 - 571 = 52 \text{ kJ mol}^{-1}$	1	M3
		<u>Alternatives:</u>		
		$\text{N}_2\text{H}_4(l) + \text{O}_2(g) \rightarrow \text{N}_2(g) + 2\text{H}_2\text{O}(l) \quad @$	1	M1
		$\Delta H_{rxn} = \sum \Delta H^\circ_{f(\text{product})} - \sum \Delta H^\circ_{f(\text{reactant})}$ $-623.0 = [0 + 2(285.5)] - [\Delta H^\circ_{f(\text{N}_2\text{H}_4)}]$ $\Delta H^\circ_{f(\text{N}_2\text{H}_4)} = 52 \text{ kJ mol}^{-1}$	1	M2
			1	M3
	(b) (i)		4	M4 M5 M6 M7

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NO	PART	SCHEME	MARKS	
		<u>Alternatives:</u> $ \begin{array}{c} \text{Zn (s)} + \text{Cl}_2 \text{ (g)} \xrightarrow{-416.0 \text{ kJ}} \text{ZnCl (s)} \\ \downarrow +126.0 \text{ kJ} \quad \downarrow +121.0 \text{ kJ} \times 2 \\ \text{Zn (g)} + 2\text{Cl (g)} \\ \downarrow +906.0 \text{ kJ} \quad \downarrow -364.0 \text{ kJ} \times 2 \\ \text{Zn}^+ \text{ (g)} \\ \downarrow +1733 \text{ kJ} \\ \text{Zn}^{2+} \text{ (g)} + 2\text{Cl}^- \text{ (g)} \xrightarrow{\text{Lattice energy}} \text{ZnCl (s)} \end{array} $	4	M4 M5 M6 M7
	(b) (ii)	$ \begin{aligned} \Delta H^\circ_{\text{lattice}} &= -416.0 -126.0 -906.0 -1733 - (121.0 \times 2) + (364.0 \times 2) \\ &= -2695 \text{ kJ mol}^{-1} \end{aligned} $	1	M8
	(b) (iii)	Lattice energy of $\text{ZnCl}_2 >$ lattice energy of KCl Zn^{2+} has higher charge compared to K^+	1 1	M9 M10
		TOTAL	10	

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3. Two standard electrode potentials are given in **TABLE 3**.

TABLE 3

Half-cell	E° / V
$\text{Co}^{2+}(\text{aq}) + 2\text{e} \rightleftharpoons \text{Co}(\text{s})$	-0.28
$\text{Zn}^{2+}(\text{aq}) + 2\text{e} \rightleftharpoons \text{Zn}(\text{s})$	-0.76

- (a) Calculate the standard potential of the cell when the two half cells in **TABLE 3** are connected.

[2 marks]

- (b) The cell reaction in (a) is allowed to continue until cell potential drops to 0.42 V. Calculate the new concentrations of Zn^{2+} and Co^{2+} .

[Given the initial concentration of Zn^{2+} and Co^{2+} are 1.0 M]

[5 marks]

NO	PART	SCHEME	MARKS																					
3	(a)	$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}} \quad @$ $= 0.28 \text{ V} - (-0.76 \text{ V})$ $= +0.48 \text{ V}$	1	M1																				
			1	M2																				
	(b)	Given initial concentration $[\text{Zn}^{2+}]$ and $[\text{Co}^{2+}] = 1.0 \text{ M}$ <table border="1" style="margin: 10px auto;"> <tr> <td></td><td colspan="4">$\text{Zn}(\text{s}) + \text{Co}^{2+}(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{Co}(\text{s})$</td></tr> <tr> <td>Initial []</td><td>-</td><td>1 M</td><td>1 M</td><td>-</td></tr> <tr> <td>Change</td><td>-</td><td>- x</td><td>+ x</td><td>-</td></tr> <tr> <td>Final []</td><td>-</td><td>1 - x</td><td>1 + x</td><td>-</td></tr> </table> Mole of electron = 2 $E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{0.0592}{n} \log \frac{[\text{Zn}^{2+}]}{[\text{Co}^{2+}]} \quad @$ $0.42 = 0.48 - \frac{0.0592}{2} \log \frac{[1+x]}{[1-x]}$ $x = 0.98$ $[\text{Zn}^{2+}] = 1.98 \text{ M}$ $[\text{Co}^{2+}] = 0.02 \text{ M}$		$\text{Zn}(\text{s}) + \text{Co}^{2+}(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{Co}(\text{s})$				Initial []	-	1 M	1 M	-	Change	-	- x	+ x	-	Final []	-	1 - x	1 + x	-	1	M3
	$\text{Zn}(\text{s}) + \text{Co}^{2+}(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{Co}(\text{s})$																							
Initial []	-	1 M	1 M	-																				
Change	-	- x	+ x	-																				
Final []	-	1 - x	1 + x	-																				
			1	M4																				
			1	M5																				
			1	M6																				
			1	M7																				
		TOTAL	7																					

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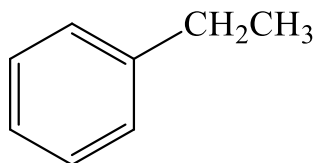
4. Hydrocarbon **A** contains one primary carbon, four secondary carbon and one tertiary carbon. This hydrocarbon reacts with bromine in the presence of light and give one major product **B**. Deduce the structural formulae of **A** and **B**.

[6 marks]

NO	PART	SCHEME	MARKS	
4		Hydrocarbon A has four – CH₂ one – CH one – CH₃ } Thus, molecular formula hydrocarbon A is C₆H₁₂ Reaction with Br₂ in the presence of light indicate hydrocarbon A is an alkane @ no carbon-carbon double bond @ saturated hydrocarbon. OR Hydrocarbon A has all sp³ hybridized carbon, thus A is an alkane @ no carbon-carbon double bond @ saturated hydrocarbon. C₆H₁₂ is general formula of alkene, however hydrocarbon A has no double bond, thus hydrocarbon A is cyclic compound. Structure A: <chem>CC1CCCC1</chem> @ <chem>CCC1CCC1</chem> @ <chem>CCC1CC1</chem> Halogenation of hydrocarbon A with Br₂ is selective, thus only tertiary hydrogen that is replaced by bromine. Structure B: <chem>CC1(Br)CCCC1</chem> @ <chem>CCC1(Br)CC1</chem> @ <chem>CCC1(Br)CC1</chem>	1	M1
			1	M2
			1	M3
			1	M4
			1	M5
			1	M6
		TOTAL	6	

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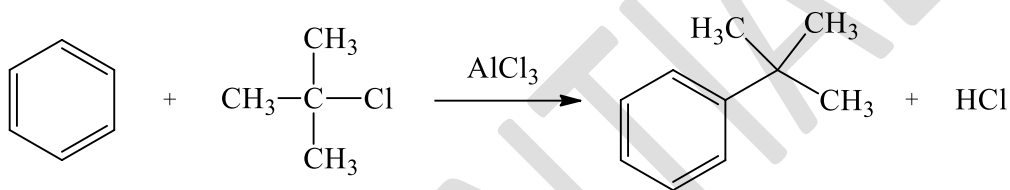
5. (a) The structural formula of compound **D** is given as below:



- (i) Give the IUPAC name for the compound **D**.
(ii) Write the chemical equation when compound **D** undergoes oxidation with hot acidified potassium dichromate solution.

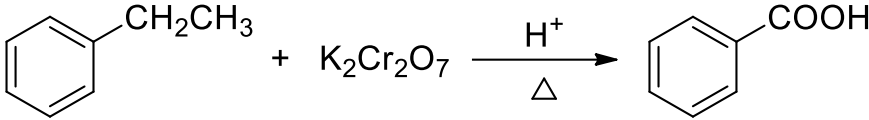
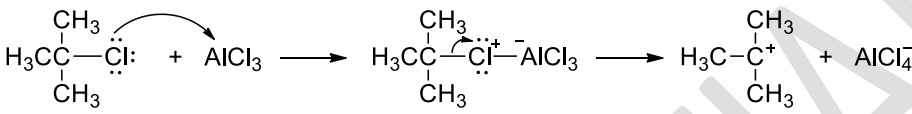
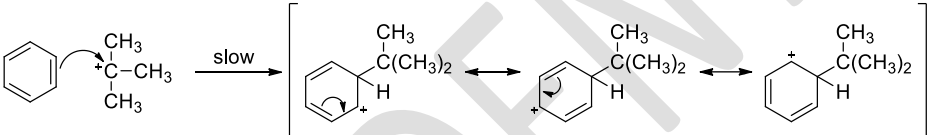
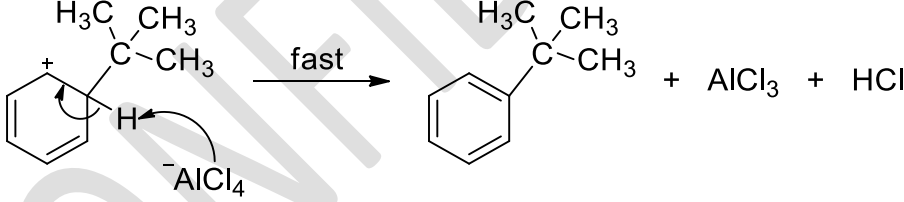
[2 marks]

- (b) Show the mechanism for the reaction below.



[5 marks]

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NO	PART	SCHEME	MARKS	
5	(a) (i)	Ethylbenzene	1	M1
	(a) (ii)		1	M2
	(b)	Step 1:  Step 2:  Step 3: 	1 (Arrow electron movement) 1 (3° carbocation) 1 (Nu-attack arrow) 1 (resonance structure) 1 (aromatisation)	M3 M4 M5 M6 M7
TOTAL			7	

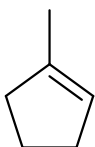
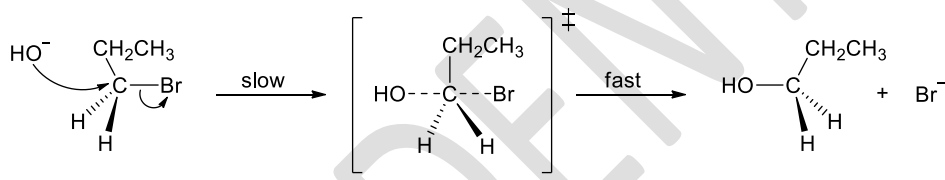
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6. (a) 2-methylbromocyclopentane reacts with KOH in ethanol to form more than one product. Draw the structural formula of the major product and explain your answer.

[2 marks]

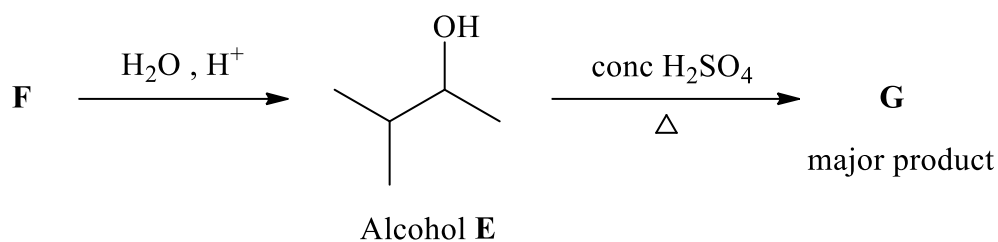
- (b) Illustrate the mechanism involved when 1-bromopropane reacts with aqueous NaOH. State the type of mechanism with explanation.

[5 marks]

NO	PART	SCHEME	MARKS	
6	(a)	 Major product satisfy Saytzeff's rule.	1	M1
			1	M2
	(b)	 S_N2 reaction because the substrate/starting material is primary alkyl halide.	1 (Nu-attack arrow) 1 (transition state) 1 (product)	M3 M4 M5
			1+1	M6 M7
TOTAL			7	

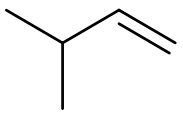
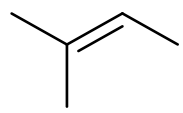
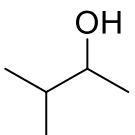
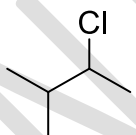
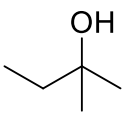
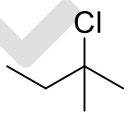
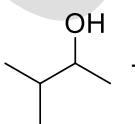
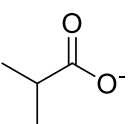
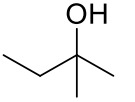
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7. Reactions involving alcohol E are shown in the scheme below.



- (a) Draw the structural formulae for **F** and **G**. [2 marks]
- (b) Give one chemical test to differentiate alcohol **E** from 2-methyl-2-butanol. State the observation and write the chemical equation. [3 marks]

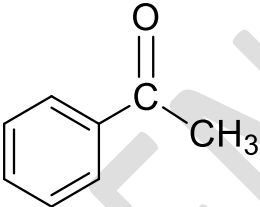
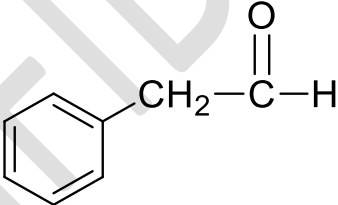
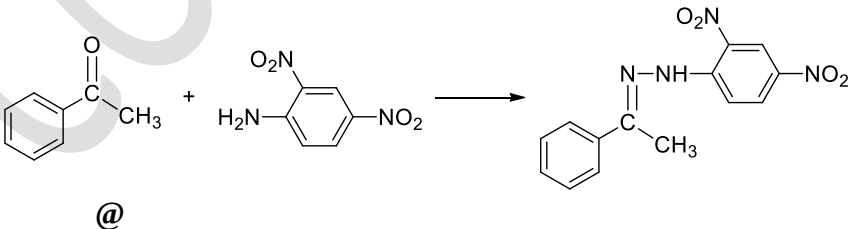
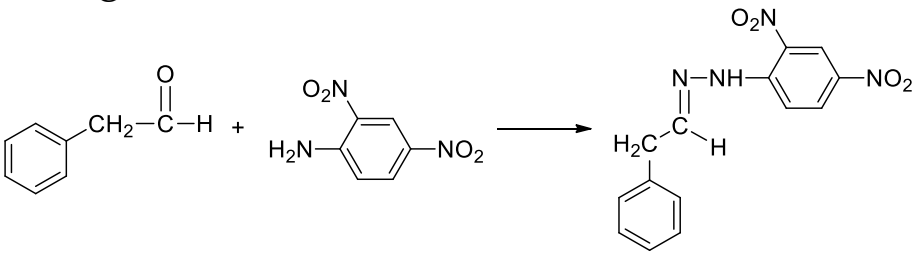
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NO	PART	SCHEME	MARKS	
7	(a)	F =  G = 	1	M1
			1	M2
	(b)	Lucas test Observation: Alcohol E form cloudy solution within 5 to 10 minutes while 2-methyl-2-butanol form cloudy solution immediately. Equation:  + conc HCl $\xrightarrow{\text{ZnCl}_2}$  + H₂O Alcohol E  + conc HCl $\xrightarrow{\text{ZnCl}_2}$  + H₂O 2-methyl-2-butanol <u>Alternative:</u> Iodoform test Observation: Alcohol E form yellow precipitate while no observable changes for 2-methyl-2-butanol. Equation:  + I₂ $\xrightarrow{\text{NaOH}}$  + CHI₃ Alcohol E  + I₂ $\xrightarrow{\text{NaOH}}$ no observable change 2-methyl-2-butanol	1	M3
			1	M4
			1	M5
			1	M3
			1	M4
			1	M5
		TOTAL	5	

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8. A monosubstituted aromatic compound with molecular formula C_8H_8O exists in two isomeric forms, **M** and **N**. Both isomers **M** and **N** gives orange precipitate with Brady's reagent. Isomer **M** produces a yellow precipitate with iodoform reagent while isomer **N** produces a silver mirror with Tollens' reagent. Deduce the structural formulae of **M** and **N**. Write all chemical equations involved.

[7 marks]

NO	PART	SCHEME	MARKS	
8		M and N gives orange precipitate with Brady's test, thus M and N has presence of carbonyl group. M gives yellow precipitate with iodoform test, hence M has presence of methyl ketone group. Structure of M =  N gives silver mirror with Tollens' reagent, thus N is an aldehyde. Structure of N:  Reaction equations:  @ 	1	M1
			1	M2
			1	M3
			1	M4
			1	M5
			1	M6

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NO	PART	SCHEME	MARKS	
		$\text{C}_6\text{H}_5\text{COCH}_3 + \text{I}_2 \xrightarrow{\text{OH}^-} \text{C}_6\text{H}_5\text{COO}^- + \text{CHI}_3$	1	M7
		$\text{C}_6\text{H}_5\text{CH}_2\text{CHO} + [\text{Ag}(\text{NH}_3)_2]^+ \xrightarrow{\text{OH}^-} \text{C}_6\text{H}_5\text{CH}_2\text{COO}^- + \text{Ag}$	1	M8
		TOTAL	8	
		MAXIMUM	7	

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9. (a) TABLE 9 below shows pKa values of three carboxylic acids.

TABLE 9

Compound	pKa
2-chloroethanoic acid, CH ₂ ClCOOH	2.86
Benzoic acid, C ₆ H ₅ COOH	4.20
Ethanoic acid, CH ₃ COOH	4.75

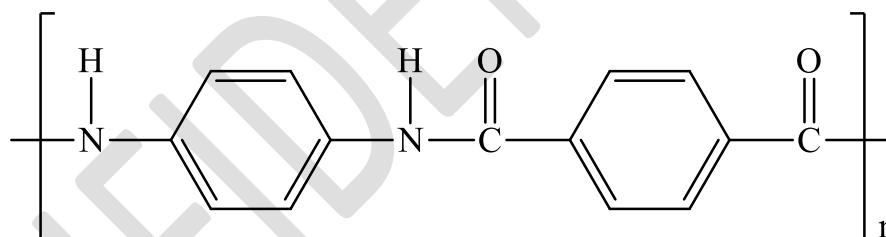
Arrange the above carboxylic acids in ascending order of acidity. Explain your answer.

[5 marks]

- (b) Draw the structural formula of product formed when propanoic acid reacts with the following reagents:
- SOCl₂
 - LiAlH₄ followed by hydrolysis.
 - ethanol in the presence of traces of mineral acid.

[3 marks]

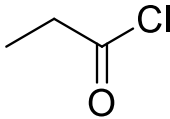
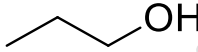
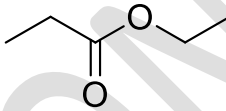
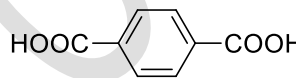
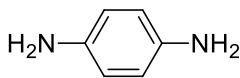
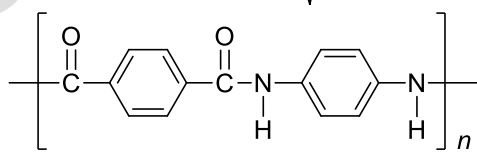
- (c) Kevlar is a very strong material and is used for making bulletproof vests. The structural formula of Kevlar is shown below.



Write the chemical equation for the formation of Kevlar through condensation polymerization.

[2 marks]

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NO	PART	SCHEME	MARKS	
9	(a)	$\text{Ethanoic acid, Benzoic acid, 2-chloroethanoic acid} \rightarrow$ Ascending order of acidity 2-chloroethanoic acid has the highest acidity due to the presence of electron withdrawing group @ Cl group, thus weaken the O-H bond in -COOH or stabilised the conjugate base. Ethanoic acid has the lowest acidity due to the presence of electron donating group @ -CH₃, thus strengthen the O-H bond in -COOH @ destabilised the conjugate base. OR Benzoic acid is more acidic than ethanoic acid because the benzoate ion is resonance stabilised, thus the delocalisation of electron in benzene ring weaken the O-H bond in -COOH @ stabilised the conjugate base.	1 1 1 1	M1 M2 M3 M4 M5 M4 M5
	(b) (i)		1	M6
	(b) (ii)		1	M7
	(b) (iii)		1	M8
	(c)	n  $+$ n  ↓  $+$ $n\text{H}_2\text{O}$	1 (Structure of two monomers) 1 (complete eq)	M9 M10
		TOTAL	10	

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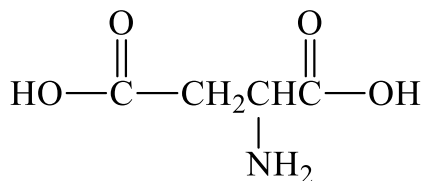
10. Compound **P** has molecular formula of C_3H_9N and exist in the isomeric forms.

- (a) Draw all structural formulae of isomers **P**. [3 marks]
- (b) Classify the isomers into primary, secondary, and tertiary amines. [3 marks]
- (c) Name a chemical test to differentiate the classes of amines. [1 mark]

NO	PART	SCHEME	MARKS	
10	(a)	$CH_3CH_2CH_2NH_2$	1	M1
		$CH_3CH(NH_2)CH_3$	1	M2
		$CH_3CH_2NHCH_3$	1	M3
		$CH_3N(CH_3)_2$	1	M4
	(b)	1° amine: $CH_3CH_2CH_2NH_2$ @ $CH_3CH(NH_2)CH_3$	1	M5
		2° amine: $CH_3CH_2NHCH_3$	1	M6
		3° amine: $CH_3N(CH_3)_2$	1	M7
	(c)	Hinsberg's Test @ Nitrous acid test	1	M8
		TOTAL	8	
		MAXIMUM	7	

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11. Aspartic acid is a type of amino acids that are used as building blocks to make protein in the body. The structural formula of aspartic acid is shown below. The isoelectric point (pI) of aspartic acid is 6.1.



Aspartic acid

- (a) Draw the structural formulae of aspartic acid at pI and pH > 8.
 (b) Draw the structural formulae of a dipeptide formed from aspartic acid. Show the peptide linkage.

[4 marks]

NO	PART	SCHEME	MARKS	
11	(a)	At pI: $\begin{array}{c} \text{H}_3\text{N}^+-\text{CH}-\text{COO}^- \\ \\ \text{CH}_2 \\ \\ \text{COOH} \end{array}$ pH > 8: $\begin{array}{c} \text{NH}_2 \\ \\ ^-\text{OOC}-\text{CH}-\text{COO}^- \end{array}$	1 1	M1 M2
	(b)	Peptide linkage	1 (Structure) 1 (label)	M3 M4
		TOTAL	4	

END OF QUESTION PAPER